C Pointers - to Medium Questions (Practice Set)

Question 1: Swap Two Numbers Using Pointers ( )

Problem Statement: Write a C program to swap two numbers using pointers.

Input Format: - Two integers a and b

Output Format: - Two lines: - Before swap: a b - After swap: a b

Constraints: - -1000 a, b 1000

Test Cases:

Test Case	Input	Output
1	`10 20`	`Before swap: 10 20After swap: 20 10`
2	`5 5`	`Before swap: 5 5After swap: 5 5`
3	`-10 15`	`Before swap: -10 15After swap: 15 -10`
4	`0 100`	`Before swap: 0 100After swap: 100 0`
5	`-50 -25`	`Before swap: -50 -25After swap: -25 -50`

Question 2: Find Maximum Element Using Pointers ( )

Problem Statement: Write a C program to find the maximum element in an array using pointer arithmetic (no array indexing []).

Input Format: - First line: integer n (size of array) - Second line: n space-separated integers

Output Format: - The maximum element

Constraints: - 1 n 100 - -1000 array elements 1000

Test Cases:

Test Case	Input	Output
1	`510 20 30 40 50`	`50`
2	`4-5 -10 -3 -8`	`-3`
3	`142`	`42`
4	`6100 50 200 75 150 25`	`200`
5	`30 0 0`	`0`

Question 3: Count Vowels in String Using Pointers ( )

Problem Statement: Write a C program to count the number of vowels (a, e, i, o, u - both uppercase and lowercase) in a string using pointers.

Input Format: - A single line containing a string (max length 100)

Output Format: - Number of vowels

Constraints: - 1 length 100 - String contains only alphabets and spaces

Test Cases:

Test Case	Input	Output
1	`hello world`	`3`
2	`AEIOU`	`5`
3	`programming`	`3`
4	`xyz`	`0`
5	`Beautiful Day`	`6`

Question 4: Copy String Using Pointers ( )

Problem Statement: Write a C program to copy one string to another using pointers without using strcpy() or any string library functions.

Input Format: - A single line containing a string (max length 100)

Output Format: - The copied string

Constraints: - 1 length 100

Test Cases:

Test Case	Input	Output
1	`hello`	`hello`
2	`C Programming`	`C Programming`
3	`12345`	`12345`
4	`a`	`a`
5	`Test String!`	`Test String!`

Question 5: Sum of Array Elements Using Pointers ( )

Problem Statement: Write a C program to calculate the sum of all elements in an array using pointer arithmetic.

Input Format: - First line: integer n (size of array) - Second line: n space-separated integers

Output Format: - Sum of all elements

Constraints: - 1 n 100 - -1000 array elements 1000

Test Cases:

Test Case	Input	Output
1	51 2 3 4 5	15
2	410 20 30 40	100
3	3-5 5 10	10
4	1100	100
5	6-10 -20 -30 10 20 30	0

Question 6: Reverse Array Using Pointers

Problem Statement: Write a C program to reverse an array in-place using two pointers (one at start, one at end) without using array indexing.

Input Format: - First line: integer n (size of array) - Second line: n space-separated integers

Output Format: - The reversed array (space-separated)

Constraints: - 1 n 100

Test Cases:

Test Case	Input	Output
1	51 2 3 4 5	5 4 3 2 1
2	410 20 30 40	40 30 20 10
3	17	7
4	6-5 10 -15 20 -25 30	30 -25 20 -15 10 -5
5	3100 200 300	300 200 100

Question 7: Check Palindrome String Using Pointers

Problem Statement: Write a C program to check if a string is a palindrome using two pointers (start and end) without using string library functions.

Input Format: - A single line containing a string (max length 100, no spaces)

Output Format: - “YES” if palindrome, “NO” otherwise

Constraints: - 1 length 100 - String contains only lowercase letters

Test Cases:

Test Case	Input	Output
1	`racecar`	`YES`
2	`hello`	`NO`
3	`madam`	`YES`
4	`a`	`YES`
5	`abcba`	`YES`

Question 8: Concatenate Two Strings Using Pointers

Problem Statement: Write a C program to concatenate two strings using pointers without using strcat() or any string library functions.

Input Format: - First line: first string - Second line: second string

Output Format: - Concatenated string

Constraints: - 1 length of each string 50 - Total length 100

Test Cases:

Test Case	Input	Output
1	`HelloWorld`	`HelloWorld`
2	`CProgramming`	`CProgramming`
3	`GoodMorning`	`GoodMorning`
4	`Test123`	`Test123`
5	`ab`	`ab`

Question 9: Find Element in Array Using Pointers

Problem Statement: Write a C program to search for an element in an array using pointers and return its position (1-indexed). If not found, return -1.

Input Format: - First line: integer n (size of array) - Second line: n space-separated integers - Third line: integer x (element to search)

Output Format: - Position of element (1-indexed) or -1 if found

Constraints: - 1 n 100 - -1000 array elements, x 1000

Test Cases:

Test Case	Input	Output
1	`510 20 30 40 5030`	`3`
2	`45 10 15 2025`	`-1`
3	`61 2 3 4 5 61`	`1`
4	`3-5 0 55`	`3`
5	`1100100`	`1`

Question 10: Count Words in String Using Pointers

Problem Statement: Write a C program to count the number of words in a string using pointers. Words are separated by spaces.

Input Format: - A single line containing a string (max length 200)

Output Format: - Number of words

Constraints: - 1 length 200 - Words are separated by single spaces - No leading or trailing spaces

Test Cases:

Test Case	Input	Output
1	`hello world`	`2`
2	`C programming is fun`	`4`
3	`single`	`1`
4	`one two three four five`	`5`
5	`the quick brown fox`	`4`
